Diamonds to power quantum computers

Scientists have developed a way to mass-produce tiny diamond crystals shaped like needles and threads, which may power the next generation of quantum computing. Physicists from Lomonosov Moscow State University in Russia have described structural peculiarities of micrometre-sized diamond crystals in needle- and thread-like shapes, and their interrelation with luminescence features and field electron emission efficiency.

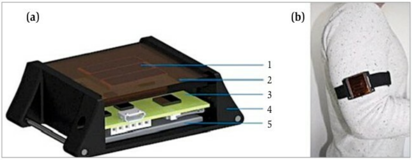
They have shown that low-quality diamond films containing separate, unconnected crystallites could be used for production of diamonds in the form of needle- or thread-like shapes. In order to achieve this, it is necessary to heat such films in an oxygen-containing environment. When heated, a part of the film material begins oxidising and gasifies. Due to the fact that diamond crystallite oxidation requires maximum temperature, it is possible to adjust the temperature so that all material except diamond crystallites is gasified.

This relatively simple technology combines production of polycrystalline diamond films with specific structural characteristics via heating in oxygen. It enables mass-production of diamond crystallites of various shapes. The crystallites could be used, for instance, as high-hardness elements, cutters for high-precision processing, or indenters or probes for scanning microscopes.

Powering medical implants with solar cells

The notion of using solar cells placed under the skin to continuously recharge implanted electronic medical devices is a viable one. Swiss researchers have found that a 3.6-square-centimetre solar cell is all that is needed to generate enough power during winter and summer seasons to power a typical pacemaker.

The study is the first to provide real-life data about the potential of using solar cells to power devices such as pacemakers and deep brain stimulators. According to lead author Lukas Bereuter of Bern University Hospital and University of Bern in Switzerland, wearing power-generating solar cells under the skin can one day save patients the discomfort of having to continuously undergo procedures to change the batteries of such life-saving devices.



(a) Cross-sectional view of the measurement device. The solar cells (1) are located directly below the optical filters (2), while the PCB (3) and the battery (5) are enclosed in the housing (4). (b) Measurement device on the upper arm (Image courtesy: www.energymatters.com.au)